

LIVERWORTS

Found mainly in damp, shaded habitats, liverworts are thought to be the simplest of all the existing groups of land plants. They come in two distinct forms: some are flat and ribbon-like; others are more like mosses, with slender stems flanked by tiny leaves.

DIVISION	MARCHANTIOPHYTA
CLASSES	3
ORDERS	16
FAMILIES	88
SPECIES	About 7,500

DEBATE PLANTS APART

Liverworts have unique features that distinguish them from mosses and hornworts, although they were once all grouped together as bryophytes. They are the only land plants without stomata, or breathing pores, in their sporophyte (spore-producing) stage. Also, their hair-like “roots” are single-celled. They are therefore classed in their own division Marchantiophyta.

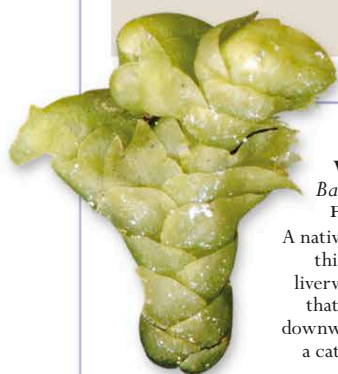
Ribbon-like liverworts look unlike any other plants. Instead of stems and leaves, they have a flat body, or thallus, that repeatedly forks as it grows. Many species have a glistening upper surface with deeply divided lobes. Medieval herbalists noted this liver-like shape, giving the plants their common name. Leafy liverworts look quite different, with trailing or spreading stems. They usually have two ranks of leaves, with a third row of smaller leaves on their underside. Some kinds form loose mats in damp grass, and others grow on rocks. They are much more numerous than ribbon-like liverworts, especially in the tropics, where many species live epiphytically on rainforest trees.

SPREADING THEIR KIND

Unlike most flowering plants, liverworts have no preset limits to their growth. They can form patches up to several metres across, which

gradually fragment as they grow, forming multiple plants. In addition, some species have cell clusters, called gemmae, in depressions on their upper surface. Raindrops scatter these gemmae, which grow into new plants.

Liverworts also reproduce by scattering spores, microscopic cells that form new plants. Spore formation requires damp conditions, because a liverwort’s female cells must first be fertilized by swimming male sperm. Sperm cells often reach egg cells with the help of raindrops, which splash them from one plant to another. Many liverworts produce their sperm or egg cells in structures that look like tiny parasols. After fertilization, the developed spores are scattered into the air.



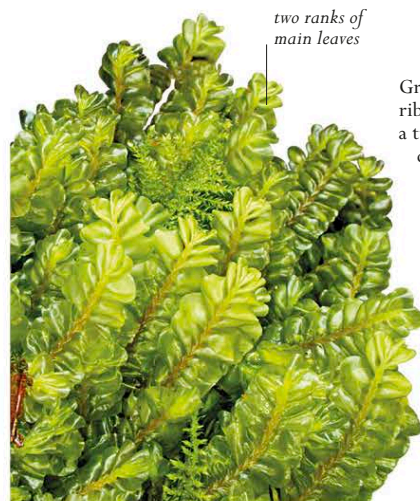
GREATER WHIPWORT
Bazzania trilobata
F: Lepidoziaceae

A native of damp woodland, this mound-forming liverwort has main leaves that overlap and curve downwards, giving its stems a caterpillar-like shape.



CRESCENT-CUP LIVERWORT
Lunularia cruciata
F: Lunulariaceae

Common in gardens and greenhouses, this ribbon-shaped liverwort is bright green, with distinctive reproductive structures, gemma cups, that look like tiny fingernails.



GREATER FEATHERWORT
Plagiochila asplenioides
F: Plagiochilaceae

A delicate liverwort with translucent leaves, this species forms spreading, moss-like tufts. It grows on shaded rocks and soil, particularly on chalk and limestone.

OVERLEAF PELLIA
Pellia epiphylla
F: Pelliaceae

Growing on wet peat and rocks, this ribbon-shaped liverwort often forms a tufted mat. It produces black spore capsules on slender white stalks.



EVEN SCALEWORT
Radula complanata
F: Radulaceae

Ranging in colour from light green to brown, this scaly-leaved liverwort creeps over flat, unshaded surfaces as varied as tree trunks and coastal rocks.



MALE REPRODUCTIVE
STRUCTURES



COMMON LIVERWORT
Marchantia polymorpha
F: Marchantiaceae

In spring and summer this ribbon-shaped liverwort produces conspicuous spore-forming structures that look like tiny parasols. It is widespread in damp habitats and gardens.